

REVERSE ENGINEERING ASSIGNMENT WRITE-UP

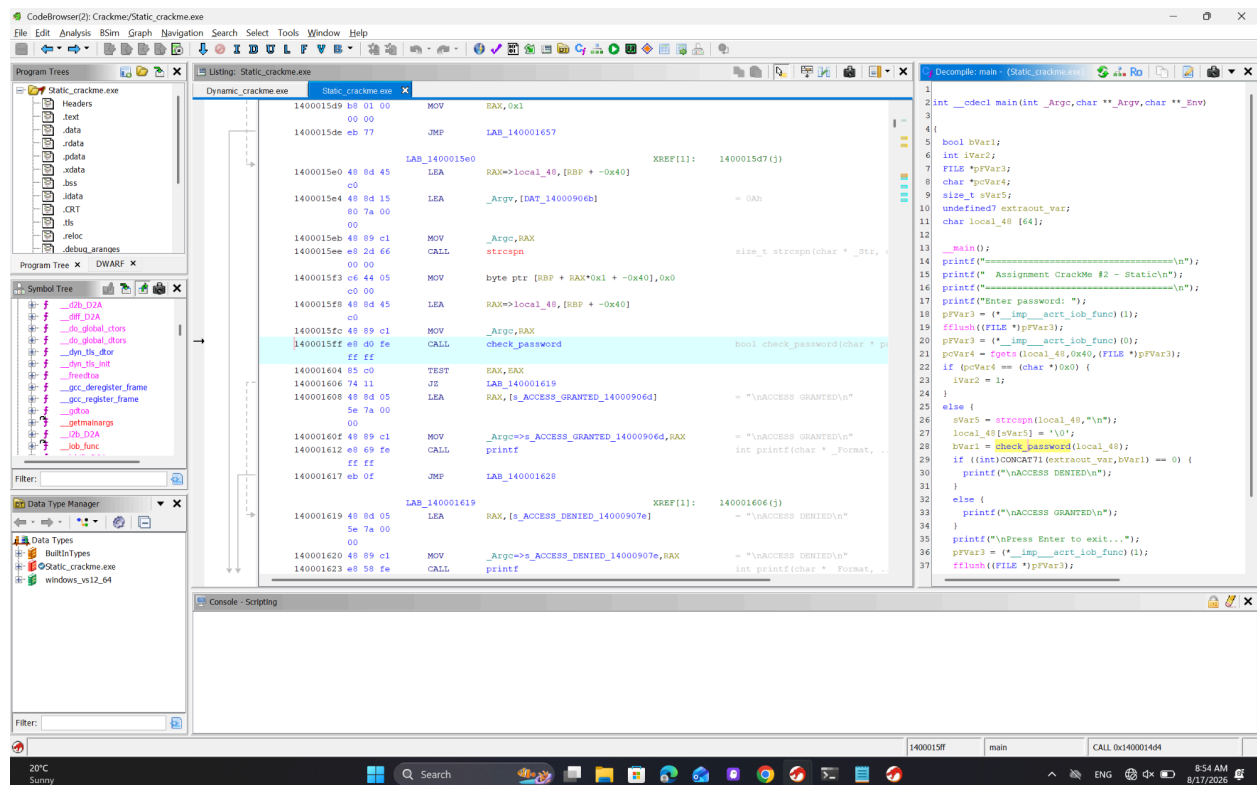
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Reverse Engineering Write-up

Reverse Engineering Assignment Write-up

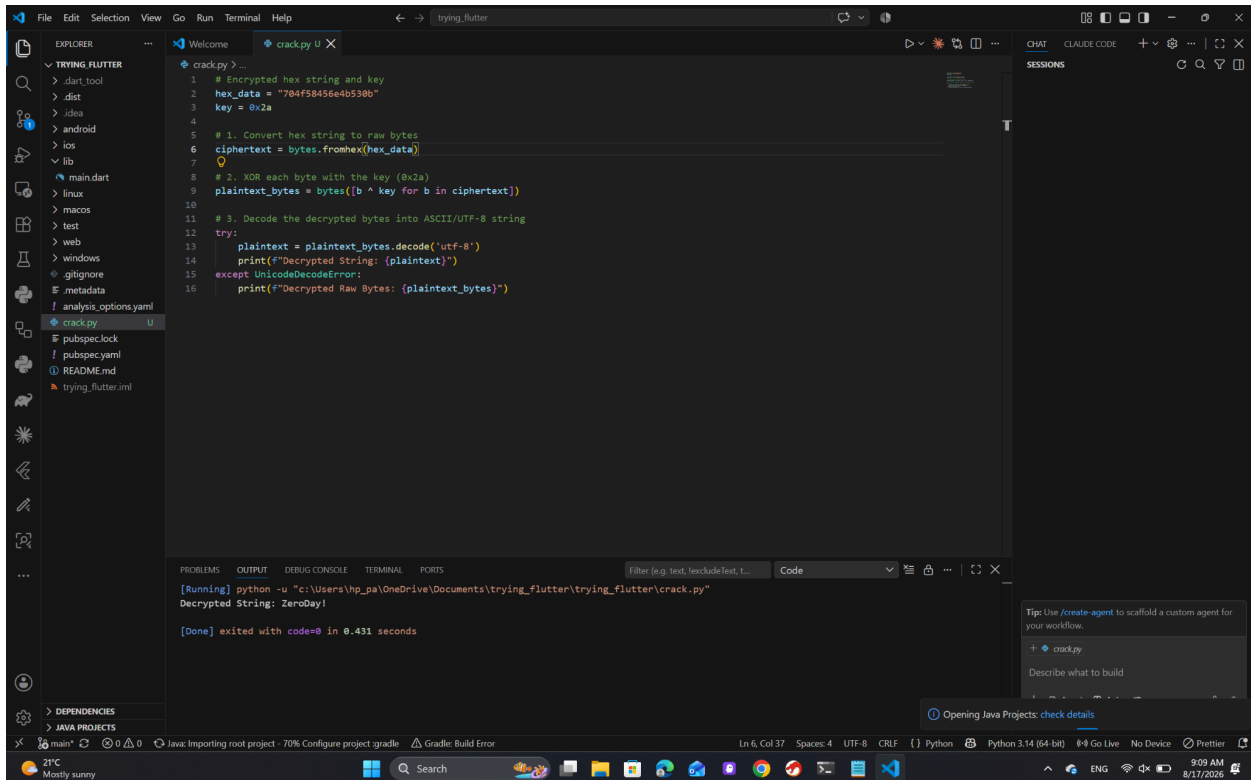
Initially I started by opening `Static_crackme.exe` in Ghidra. After the file was analyzed, I went to the **Functions** section and searched for the `_main` function. From there, I could see the code related to the password check.



Screenshot 1: Ghidra showing the `_main` function and `check_password`.

I then opened the `check_password` function. I found `DAT_140009000`, and when I followed it, I saw the XOR operation and the encryption key.

I used a small Python script in VS Code to decrypt the value.



The screenshot shows the Visual Studio Code interface. The Explorer sidebar on the left shows a project named 'TRYING_FLUTTER' with various files and folders. The main editor area displays a Python script named 'crack.py'. The script defines a hex string 'hex_data' and a key 'key', then performs XOR decryption. The output of the script is shown in the terminal at the bottom, which reads 'Decrypted String: ZeroDay!'. The terminal also shows the command used to run the script: 'python -u "c:\Users\hp_pa\OneDrive\Documents\trying_flutter\trying_flutter\crack.py"'.

```
1 # Encrypted hex string and key
2 hex_data = "704f5845e4b530b"
3 key = 0x2a
4
5 # 1. Convert hex string to raw bytes
6 ciphertext = bytes.fromhex(hex_data)
7
8 # 2. XOR each byte with the key (0x2a)
9 plaintext_bytes = bytes([b ^ key for b in ciphertext])
10
11 # 3. Decode the decrypted bytes into ASCII/UTF-8 string
12 try:
13     plaintext = plaintext_bytes.decode('utf-8')
14     print(f"Decrypted String: {plaintext}")
15 except UnicodeDecodeError:
16     print(f"Decrypted Raw Bytes: {plaintext_bytes}")
```

```
[Running] python -u "c:\Users\hp_pa\OneDrive\Documents\trying_flutter\trying_flutter\crack.py"
Decrypted String: ZeroDay!

[Done] exited with code=0 in 0.431 seconds
```

Screenshot 2: Python code in VS Code and the output .

Decrypted String : ZeroDay!

After getting the password, I ran `Static_crackme.exe` from CMD and entered the password. The program accepted it and showed **Access Granted**.

```
PS C:\Users\hp_pa\OneDrive\Documents\INSA Summer camp\reverse\crackmes> .\Static_crackme.exe
=====
Assignment CrackMe #2 - Static
=====
Enter password:

ACCESS DENIED

Press Enter to exit...
PS C:\Users\hp_pa\OneDrive\Documents\INSA Summer camp\reverse\crackmes> .\Static_crackme.exe
=====
Assignment CrackMe #2 - Static
=====
Enter password: ZeroDay!

ACCESS DENIED

Press Enter to exit...
PS C:\Users\hp_pa\OneDrive\Documents\INSA Summer camp\reverse\crackmes> .\Static_crackme.exe
=====
Assignment CrackMe #2 - Static
=====
Enter password: ZeroDay!

ACCESS GRANTED

Press Enter to exit...|
```

Screenshot 3: The password entered in CMD and the Access Granted result.

Result

I was able to find the encrypted password in Ghidra, decrypt it using the XOR key with Python, and confirm that the password worked by testing it in the executable.